1. Reading Comprehension

How Scientists Read: The Hidden Skill Behind Discovery

Scientific reading is a skill that distinguishes a passive learner from an active researcher. In every scientific field – from molecular biology to astrophysics – the ability to understand and evaluate research articles determines how successfully one can learn, question, and innovate. Yet, many university students approach scientific texts as if they were reading a newspaper. They expect immediate clarity, simple sentences, and personal opinions. In reality, research writing demands much more: attention to structure, precision in terminology, and a critical mindset.

A scientific article typically follows a standard format known as **IMRaD**: *Introduction, Methods, Results, and Discussion.* Each section serves a unique purpose. The *Introduction* presents the background and defines the problem; the *Methods* explain how the study was conducted; the *Results* describe what was found; and the *Discussion* interprets these findings, connecting them to existing knowledge. Understanding this structure helps readers predict what type of information they will encounter, which in turn improves comprehension.

However, the challenge lies not only in structure but also in **language**. Research articles often employ complex syntax, nominalized verbs (e.g., "observation" instead of "observe"), and passive constructions that emphasize the process rather than the person. This formal tone creates objectivity but can make the text seem distant or abstract. For instance, "The samples were collected from three sites" hides the human agent, focusing solely on the scientific act. Such language choices are intentional – they reflect the discipline's values of precision, neutrality, and replicability.

Moreover, **reading critically** is as important as understanding the words. Critical readers question the author's assumptions, compare data with other studies, and evaluate whether the conclusions are supported by evidence. Blind acceptance of scientific claims can be as dangerous as ignorance itself. As one researcher once said, "Good scientists doubt everything, including their own results." This intellectual humility drives progress, since science evolves through correction, debate, and reinterpretation.

For Moroccan science students, engaging with English-language research has double value: it builds both linguistic competence and professional awareness. Around 80% of global scientific publications appear in English. Therefore, students who master scientific reading not only access the latest findings but also gain the vocabulary and analytical thinking necessary for postgraduate research or international collaboration. Scientific literacy, ultimately, is not about reading more – it is about reading *better*.

Comprehension and Critical Thinking Tasks

A. Identifying Structure

1. The text mentions the "IMRaD" structure. Match each letter with its correct description:
   - I = _____
   - M = _____
   - R = _____
   - D = _____
2. Why is recognizing this structure helpful when reading a research article?

B. Vocabulary in Context

Find words or expressions in the text that mean the following:

1. The quality of being exact and detailed: _____
2. To think carefully and form a judgment: _____
3. Words made from verbs to form abstract nouns (e.g., "observation"): _____
4. The act of checking whether something can be repeated or confirmed: _____

C. Cause and Effect

Complete the sentences using information from the text.

1. Research writing uses passive voice because _____.
2. Students who read research articles in English benefit because _____.
3. Critical reading improves science because _____.

D. Paraphrase Recognition

Choose the sentence that best expresses the same idea as the one in the text.

1. *"Good scientists doubt everything, including their own results."*
   a) Scientists should always trust their first findings.
   ✗ b) Skepticism is part of scientific progress.
   c) Scientists must never question their own methods.
   d) Doubt prevents discovery.
2. *"Scientific literacy is not about reading more – it is about reading better."*
   a) Quantity of reading is more important than quality.
   b) Students should read everything available.
   ✗ c) Deep comprehension matters more than volume.
   d) Reading less always leads to success.

E. Sequencing Ideas

Arrange these ideas in the order they appear in the text (1 – 5):

5 The importance of English in global science
2 The structure of scientific papers
3 The formal language used in research writing
4 The role of critical reading in science
1 The difference between passive and active learners

F. Discussion Questions (Oral / Written)

1. Why do scientific texts use complex grammar and vocabulary instead of simple expressions?
2. In what ways does critical reading help prevent misinformation in science?
3. Do you agree that reading "better" is more valuable than reading "more"? Explain.

2. Vocabulary Development

A. Academic Word List

Word	Part of Speech	Definition	Example Sentence
precision	noun	exactness, accuracy	Scientific instruments must operate with great precision.
assumption	noun	a belief accepted without proof	Every hypothesis begins with an assumption.
interpret	verb	explain or understand the meaning	Students must interpret data before drawing conclusions.
neutrality	noun	the state of not showing bias	Neutrality is essential when analyzing research.
replicate	verb	to repeat an experiment to verify results	Other labs tried to replicate the experiment.

B. Word Formation – Complete the table.

Verb	Noun	Adjective
interpret	interpretation	interpretative
assume	assumption	
analyze	_____	_____
observe	_____	_____
innovate	_____	_____

3. Grammar: Relative Clauses

A. Explanation

A **relative clause** gives extra information about a noun.
It is introduced by a relative pronoun: **who, which, that, whose, where.**

Pronoun	Used For	Example
who	people	The scientist **who** led the study won an award.
which	things	The method **which** was used produced reliable results.
that	people or things	The article **that** you read is from *Nature*.
whose	possession	The student **whose** paper was selected received a grant.
where	place	The lab **where** the samples were analyzed is in Casablanca.

B. Practice 1 – Combine Sentences

1. The professor explained the experiment. He wrote the textbook we use.
   → The professor **who** _____.
2. The study was published last month. It examines renewable energy.
   → The study **that** _____.
3. The university built a new laboratory. Students can test chemical reactions there.
   → The university built a new laboratory **where** _____.
4. The researcher's data were precise. Her accuracy impressed the committee.
   → The researcher **whose** _____.

C. Practice 2 – Identify and Correct

Find and correct the error in each sentence.

1. The microscope which we buyed yesterday is broken.
2. The scientist who's theory was wrong corrected it later.
3. The journal that publish the article is based in London.

D. Practice 3 – Complete Sentences

Use relative clauses to make sentences more informative.

1. Science is a discipline _____.
2. A hypothesis is an idea _____.
3. A scientist is someone _____.
4. The data _____ were difficult to interpret.

4. Speaking

Group Task:
Each group chooses a short research article (in English) from a scientific journal or online source.

- Summarize the main idea in 2 – 3 minutes.
- Use at least **three relative clauses** and **five key vocabulary items** from today's lesson.
- Discuss: *What makes this article reliable or unreliable?*

5. Writing

A **summary** is a **short version** of a text that presents only the **main ideas** in your own words.
Follow these steps:

1. **Read** the text carefully and identify the **main argument**.
2. **Highlight** key points, not examples or statistics.
3. **Avoid copying sentences**; paraphrase the author's ideas.
4. **Keep it objective** – do not add opinions or comments.
5. **Use connectors** (e.g., firstly, moreover, finally) and concise language.
6. **Check** that your summary is coherent and clear.

Write a **200-word analytical summary** of the text *"How Scientists Read: The Hidden Skill Behind Discovery."*

Your summary should:

- Identify the text's **main argument and structure (IMRaD)**
- Include **three relative clauses**
- Use **vocabulary** such as *assumption, interpret, precision, replicate,* and *neutrality*
- Conclude with your opinion on why critical reading is essential for scientific success